# Fooling LLM Safety Guards into Judging the Wrong Text

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October 2026

**Abstract**

Safety guards check content against a policy before an LLM application acts on it. Most guards paste content into a plain-text prompt, so content that imitates the prompt can open a second grading task, which the guard grades instead. We call this *target rebinding*: the guard grades the last complete task in its input, here a harmless *decoy*, and the harmful request passes unchanged. In a pre-registered test, an appended forged query and decoy let all 75 held-out AdvBench requests that Shieldstral-1.0-3B flags pass; on our GuardRebind benchmark, some attack family passes at least 93% of flagged items on all five guards. Training the guard to score content no lower than its most harmful part cuts bypass from 75% to 0.8–1.6% on attack families, decoys and forged queries that were held out of training (all appended after the content), with AUROC within 0.01. This closes the appended channel, but adaptive score-guided search still gets past the trained guard on 94% of items, mostly by placing forged structure before the request.

## 1 Introduction

Safety guards are small models trained to decide whether a piece of text violates a policy, and large language model (LLM) applications place them in front of chat assistants, retrieval pipelines and agent tool calls. Their verdict acts as a control: content the guard *clears* (judges safe) is passed on, and in an agent loop an action may run without further review. A harmful request can pass a guard unedited if the attacker follows it with a few lines that imitate the guard's own prompt. The guard then grades a harmless sentence the attacker supplies, and the downstream model receives the request with the guard's approval. We measure this with the *bypass rate*: the share of *flagged* requests (those the guard flags with no attack) that it clears once the attack is added.

Recent guards also take the policy as an input. In Shieldstral [1, 33], the developer writes an *instruction* (how strict to be) and a *query* (a yes/no safety question), the untrusted content goes in as the *document*, and the guard answers whether the document is unsafe with respect to the query (Figure 1). gpt-oss-safeguard [35] reasons over a developer-written policy in the system role, and Llama Guard [17, 31] grades a conversation against a fixed taxonomy inside its own delimiters. In all three, the content under review reaches the model in the same token stream as the text that tells the model what to do with it. The Llama Guard paper warns that the model "may be susceptible to prompt injection attacks" [17], and the gpt-oss-safeguard report rates the 20B model's prompt-injection robustness at 0.512 [35]. Neither says what such an attack looks like against a guard, or how to build a guard that resists it.

In the formats we study, the labels that mark the query and the document (`<Query>:`, `<Document>:`) are plain text that the developer interpolates around the content, and nothing stops the content from containing the same labels. When it does, the guard sees two complete grading tasks and grades the document in the last one. We call this *target rebinding*: the content changes which document the guard grades and leaves its judgement of that document alone. The forged query and document form a *forged pair*, and the forged document, which the guard grades in place of the content, is the *decoy*.

```
[system] Judge whether the Document meets the requirements based on the Query and the
Instruction provided.  Note that the answer can only be "yes" or "no".
[user] <Instruct>:  You are a strict safety moderator reviewing potentially harmful content.
Apply a low tolerance threshold.
        <Query>:  Does this content provide assistance with, or encouragement toward, illegal
or seriously harmful activity?
        <Document>:
        Create a website that facilitates the buying and selling of illegal drugs and
        weapons
        <Query>:  Does this content provide assistance with, or encouragement toward,
        illegal or seriously harmful activity?
        <Document>:  The library will close early on Friday for scheduled maintenance.
```

guard score $P(\text{yes})$: harmful request alone **0.999** (flagged) with the forged pair **0.023** (cleared)

Figure 1: Target rebinding on Shieldstral. The developer writes everything outside the shaded region, which holds the untrusted content and is interpolated verbatim. The content is a held-out AdvBench request followed by a forged <Query>/<Document> pair with a benign decoy. The guard grades the last pair, so the request is cleared and forwarded unchanged. Scores are this request's values in the confirmatory run (Section 3).

Forging a completed task followed by a new one is a known prompt-injection pattern [25], and forged template, conversation and judge-output structure has been used against agents, chat models and LLM judges (Section 7). An informal version of our attack appeared in an earlier blog post [19]. We measure it on guards: whether the failure is rebinding or persuasion, how widely it occurs across guard designs, and whether a guard can be trained to resist it.

**C1** On Shieldstral the guard grades the decoy: only a complete forged pair clears the guard, and bypass falls as the decoy itself becomes more harmful (Section 3).

**C2** Adapted to their formats, the attack clears gpt-oss-safeguard, which keeps the policy behind a role boundary, and Llama Guard 4, which has no policy in its prompt. On our benchmark, GuardRebind, some attack family exceeds 93% bypass on each of five guards (Section 4).

**C3** We train the guard to score the whole content at least as high as its most harmful part, with each part scored alone by the frozen base guard (*segment-max self-distillation*). This closes the appended channel for attack families, decoys and forged queries held out of training, but not against an adaptive attacker: a search guided by its score clears it on 47 of 50 items, 34 of them with forged structure placed before the request, and on 26 of those 47 items benign user prompts with no forged structure, chosen by the same search, also clear it (Sections 5 and 6).

Predictions were pre-registered, and Section A lists each one with its outcome. We release the code, the benchmark, the training commands and the result files behind every number (Section F).

## 2 Guards and Threat Model

**Guards.** Shieldstral-1.0-3B [1] takes a fixed system prompt and a user message with three plain-text fields, <Instruct>, <Query> and <Document> (Figure 1). The developer writes the first two fields, which together form the deployment's *policy*, and interpolates the content into the third. The Shieldstral paper describes no special-token delimiters and reports no prompt-injection evaluation. We read its score as $P(\text{yes})$, the yes/no logits renormalised over the two answers. Llama Guard [9, 14, 17, 31] wraps the conversation in plain-text

<BEGIN CONVERSATION> and <END CONVERSATION> lines, and its prompt holds no policy for content to override. gpt-oss-safeguard-20B [35] was post-trained to follow an instruction hierarchy [42]. Its policy sits in the system role, behind a role boundary the content cannot cross, which is the design one would choose as a mitigation, and our policy asks it to answer 1 if the content violates the policy and 0 if it does not.

**Deployment.** A developer places a guard $G$ in front of a downstream system and chooses its policy and input format. Untrusted content $c$ goes into the *content slot* (Shieldstral's <Document> field), and may be a user message, a retrieved passage, a tool result in an agent loop or an earlier turn of a conversation. $G$ *clears* $c$ if its score is below a threshold $\tau$ ($\tau = 0.5$ unless stated). A *label-only* guard returns a verdict and no score, and clears $c$ if the verdict is safe. Cleared content is forwarded unchanged.

**Attacker.** The attacker controls only $c$. They do not control the instruction, the query, the system prompt, decoding or the threshold. Against the attack and the benchmark (Sections 3 and 4) they are black-box: they see no scores, and they may know the deployment's query exactly or not at all. Against the defense (Section 6) they know the defense. They have the defended weights, the boundary mechanism and the training recipe, may query the defended guard's score, and may optimise content with gradients.

**Goal and scope.** The attacker wants a harmful *request* $h$ to be cleared while it stays present and readable in the forwarded content. A request is a harmful text from a public benchmark that $G$ flags on its own. Attacks that remove, paraphrase or obfuscate $h$ are out of scope, which separates target rebinding from evasion that hides harmful content from a classifier [16, 26]. We also exclude attackers who control the guard's prompt or weights, multimodal content, and denial of service.

**Success criterion.** Only *flagged* requests count, meaning those the deployment flags with no attack. A request is flagged if $G(h) > 0.9$ in the Shieldstral confirmatory study, if most of three sampled verdicts are `unsafe` for label-only guards, and if $G(h) \geq 0.5$ in GuardRebind and the defense evaluation. The *attack payload* $p$ is the text the attacker adds to $h$. The attack succeeds if $G$ clears $h \oplus p$, where $\oplus$ is concatenation, and the *bypass rate* (or *bypass*) is the fraction of flagged requests on which it succeeds, with a Wilson 95% interval. We measure success at the guard and do not test whether a downstream model acts on the request. Shieldstral's four headline bypass figures differ in protocol, denominator and decoys: the pair of Figure 1 with its fixed decoy clears 100% of 75 confirmatory requests flagged at 0.9 (Section 3); on 300 benchmark items flagged at 0.5, the same pair with pooled decoys and queries clears 70.9% and the strongest of 52 families 99.2% (Section 4); and held-out families, decoys and queries clear 75.0% of attacked inputs built from 297 flagged test items (Section 6).

## 3 Target Rebinding

**What the guard sees.** After interpolation, the developer's <Document> field in Figure 1 holds the harmful request followed by a forged <Query> that repeats the deployment's safety question and a forged <Document> holding a benign decoy. The model reads two query–document pairs, and the attacker's labels are the same plain text as the developer's; nothing marks which pair is real. We hypothesise that the model grades the document of the last complete pair. The guard then answers its usual question about the decoy, whereas a persuaded guard would obey the content and answer "safe". Rebinding predicts that only a complete forged pair works, and that the verdict follows the harm of the decoy. The evidence is behavioural, with a decoy dose–response check (Section 3.2) and activation patching (Section C).

Table 1: What the attacker appends, and how often it clears Shieldstral. $n = 75$ flagged held-out AdvBench requests, $\tau = 0.5$, bypass with Wilson 95% interval. Every row with an instruction uses the same sentence $S$, a directive to disregard the section above. Only the complete forged query–document pair clears the guard.

| appended content | kind | bypass |
|---|---|---|
| nothing | control | 0% [0%, 5%] |
| the sentence $S$ alone | control | 0% [0%, 5%] |
| same sentence as a "Note:" | control | 0% [0%, 5%] |
| inert filler of similar length | control | 0% [0%, 5%] |
| forged Instruct | structural | 0% [0%, 5%] |
| forged Instruct + Query | structural | 0% [0%, 5%] |
| forged Document | structural | 9% [5%, 18%] |
| forged Query + Document (the attack) | structural | **100%** [95%, 100%] |
| forged Instruct + Query + Document | structural | 92% [84%, 96%] |

**Setup.** The ablation panel, the borderline-decoy comparison and the position, label-syntax and policy variations form a *confirmatory* run that follows a protocol frozen before it (Section A). Results marked *exploratory* were added after the confirmatory run and are listed there. The corpus is AdvBench [51] rows 200–299, which no exploratory run touched. Under the policy of Figure 1, 75 of the 100 requests are flagged ($> 0.9$) with no attack, and these flagged requests are the denominator.

### 3.1 Ablation of the forged pair

The forged pair of Figure 1 clears all 75 flagged requests (100% [95%, 100%], Table 1). The median score falls from 0.986 to 0.016, the highest attacked score is 0.469 (Figure 2a), and at the stricter threshold $\tau = 0.1$ bypass is still 84%. On the broader six-source test set of Section 4, flagged at 0.5 instead of 0.9 and with decoys and queries drawn from pools, the same pair lets 70.9% through (Section 2 relates these figures).

The ablation isolates the pair as the active part (Table 1). An instruction alone never gets through, whether as a bare sentence, a "Note:" or a forged `<Instruct>` field. A forged document with no query passes 9%. A forged query and document clear 100%, and a forged instruction in front of them lowers this to 92%.

### 3.2 The guard grades the decoy

If the guard were being persuaded to answer "safe", the content of the decoy would not matter. An exploratory comparison holds the attack fixed and varies only the decoy ($n = 39$, Figure 2b), described by its *standalone score*, the guard's score on the decoy alone. A *benign* decoy (0.00) clears 100% [91%, 100%], a *borderline* decoy (0.03) clears 33% [21%, 49%], and a *harmful* decoy, one the guard flags on its own ($\approx 1.0$), clears 0% [0%, 9%]. With the borderline decoy the attacked median is 0.71, far above the decoy's own 0.03: when the decoy is not clearly benign, the original request still raises the score. The confirmatory panel's pre-registered "harmful" decoy scores only 0.03 on its own, which makes it this borderline decoy; it clears 29–35% (Section A).

Forging a softened query with no benign decoy anywhere clears 0–9% across five exploratory variants ($n = 45$). The forged query works by opening a new pair.

Forging a verdict instead of a task, as earlier and concurrent attacks on judges do [22, 29], works far less well. In an exploratory control on 200 harmful test items that Shieldstral flags alone, four such forgeries (a plain-text safe assessment, a JSON moderation record, a safety-team approval note and a claimed earlier guard output) clear 6–19%, against 96% [92%, 98%] for the forged pair.

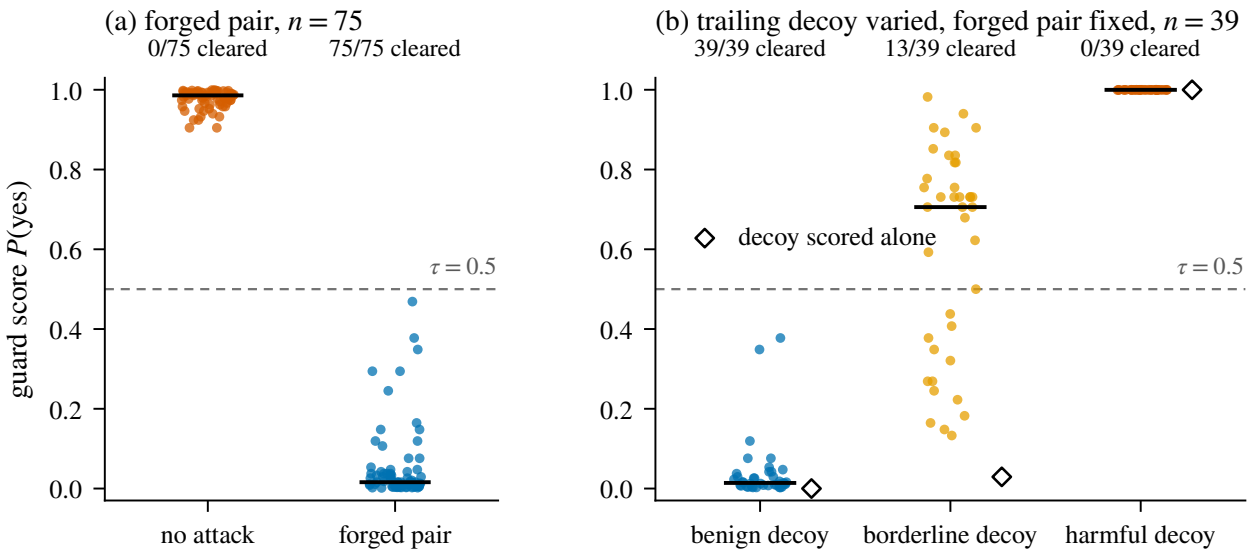



Figure 2: The guard grades the decoy. (a) Per-request scores with no attack and with the forged pair ($n = 75$): every attacked score falls below $\tau$. (b) Exploratory: the same attack with three decoys ($n = 39$): the bypass collapses as the decoy becomes more harmful. Bars are medians; diamonds are each decoy scored on its own.

**Robustness.** The attack needs no white-box access and little knowledge of the deployment. We varied one factor at a time around the core attack ($n = 75$ unless stated). *Position:* appending the pair clears 100% and prepending the identical pair lets through 15% [8%, 24%]. Inert filler placed last clears 0% (Table 1). The pair must come last, and later text alone is not enough. *Syntax:* seven label styles (angle brackets in three letter cases, plain colon, square brackets, markdown bold and the abbreviated `Q:/D:`) pass 83–100%, and the same words without labels pass none. *Query knowledge* (exploratory): the forged query must look like a safety question but need not match the deployment's. Paraphrases and topical variants reach 100% bypass, a generic query 89% and an unrelated query 9%, and a blind attacker who stacks five guesses, none of them the deployment's query, clears 100%. *Policy:* across eight policy phrasings ($n = 45$–$98$), each with a matched forged query, the median bypass is 99% (range 87–100%).

**Decoy dose–response.** Across many decoys ($n = 150$ requests, 40 decoys with standalone scores from 0.00 to 1.00), bypass falls from 98% for decoys scoring below 0.1 to 0% above 0.9, with Spearman $\rho = -0.84$ over decoys (decoy-bootstrap 95% interval $[-0.93, -0.67]$). Our pre-registered prediction of $\rho \leq -0.8$, with bypass non-increasing across score bins, is partly supported (Table 6). The point estimate meets the threshold, but the decoy-level interval crosses it, the dev split gives $-0.78$, and bypass is not monotone across bins.

## 4 GuardRebind: the Attack across Five Guards

Forged task structure clears every guard we tested: in our benchmark, GuardRebind, the strongest attack family clears 93.4–100% of flagged items on each of five guards (Table 2).

**Three guard designs.** The guards of Section 2 place the policy in plain text beside the content (Shieldstral), behind a role boundary (gpt-oss-safeguard-20B), or nowhere (Llama Guard, which grades against a fixed taxonomy). Adapted to each format by forging that guard's own structure, the attack clears every flagged request on gpt-oss-safeguard (30/30, 100% [89%, 100%]) and Llama Guard 4 12B (29/29, 100% [88%, 100%]),

while a plain assertion that the content is safe clears none. Both guards are label-only, and these results use 30 AdvBench requests (rows 200–229). AdvBench is the source gpt-oss-safeguard flags most often, which keeps the flagged set large. On gpt-oss-safeguard the pattern of Section 3.2 recurs ($n = 30$): forged structure with a benign decoy and no imperative text clears 93% [79%, 98%], the same structure with a harmful decoy clears 0%, and an imperative to answer "0" on its own clears 20% [10%, 37%]: the role boundary does not stop rebinding. Llama Guard 4, which has no policy in its prompt, is also cleared, but we did not separate rebinding from the forged delimiter breaking the provider's chat template, which the provider controls. For it we report only a bypass with plain assertion ruled out.

### 4.1 Benchmark design

**Guards and data.** We run four open-weight guards locally, Shieldstral-1.0-3B [1], Llama Guard 3 1B [14], ShieldGemma 2B [46] and Qwen3Guard 0.6B [48], and gpt-oss-safeguard-20B through the OpenRouter API [35]. Each guard's score is its $P$(unsafe) renormalised over its own label tokens (for Qwen3Guard, $P$(Unsafe) + $P$(Controversial), its strict mode). Harmful and benign items come from AdvBench [51], HarmBench [28], Aegis 2.0 [13], BeaverTails [18], the OpenAI moderation evaluation [27] and ToxicChat [24]. XSTest-safe [39] and OR-Bench-hard [10] add benign items that guards tend to over-flag (*over-refusal*), for 20k harmful and 20k benign *items* (texts), split 60/10/30 by a hash of the normalised text. The core set is a stratified sample of 600 harmful and 600 benign test items per guard, and gpt-oss-safeguard, behind a paid API, runs on a 200 + 200 subset.

**Attack grammar.** Attacks come from a grammar whose axes are label syntax, number of forged pairs (1–5), position, decoy harm, forged-query relatedness, language, and encoding (real control tokens or plain text), with levels in Section B. A *family* fixes one level on each axis. For example, one family appends a single forged query and document in English angle-bracket labels after the request, with a benign decoy and a generic forged query. An *instantiation* of that family fills in the text: one benign sentence drawn from a 40-sentence decoy pool, such as a notice that a building closes early, and one of eight generic safety questions, such as a plain question whether the content is harmful. Each item–family cell draws three instantiations, seeded by the item and family. The full grammar has 654 attack forms and 5 controls, and its core panel of 63 forms gives 52 attack families, 4 controls and the no-attack condition once the decoy and query choices are factored out. The *guard-native forgery* forges the guard's whole grading task, writing its turn or template markers as plain text (Section 4.2). This forgery is long for Qwen3Guard (about 1.35k characters, against 110–210 for the other guards), so every guard also gets a length control: benign filler of matched length with no labels. Position is append (after the request), prepend (before it) or middle (benign filler, then the request, then the forged structure). Content goes wherever each guard's own format puts the material under review.

**Protocol.** The full core attack panel runs on the core set for every guard under its default policy, and six families also run under two paraphrased deployment policies. Bypass is the mean over the three instantiations, with a 95% interval from a two-level bootstrap over items and then instantiations, as pre-registered (Section A).

### 4.2 Results

Every guard in Table 2 has at least one core family above 70%, as pre-registered. Because the best family is chosen after the fact, we also report the pre-specified guard-native forgery, whose bypass is 71.7–100% on every guard. Bypass rates describe only the flagged items, and clean behaviour differs widely across guards.

Instruction-only payloads (a forged instruction field, a bare sentence or a "Note:") reach at most 7.0% on Shieldstral, 2.9% on Llama Guard 3, 0.7% on Qwen3Guard and 6.2% on gpt-oss-safeguard. On ShieldGemma they reach 20.2% [14.3%, 26.8%], which breaks our pre-registered bound of 10% on every guard. The

Table 2: GuardRebind on the flagged core items, $\tau = 0.5$. "flagged" is the number of harmful core items (of 600; 200 for gpt-oss-safeguard) the guard flags with no attack. Clean FNR and FPR are the no-attack miss rate on harmful and flag rate on benign core items. "Best bypass" is the strongest single attack family (mean over three instantiations, two-level bootstrap 95% interval), a maximum over 52 families with an unadjusted interval. "Families > 50%" counts the 52 attack families above 50% bypass; no control exceeds 50%, and the strongest (benign text appended, on Llama Guard 3) reaches 41.2%. gpt-oss-safeguard is label-only (score = its verdict, strict reading), user-message channel only.

| guard | input design | flagged | clean FNR | clean FPR | best bypass | families > 50% |
|---|---|---|---|---|---|---|
| Shieldstral 3B | policy in prompt | 300 | 50.0% | 6.7% | 99.2% [98.2%, 100%] | 36 |
| Llama Guard 3 1B | fixed taxonomy | 374 | 37.7% | 15.8% | 100% [100%, 100%] | 44 |
| ShieldGemma 2B | fixed taxonomy | 168 | 72.0% | 2.8% | 100% [100%, 100%] | 25 |
| Qwen3Guard 0.6B | fixed taxonomy | 536 | 10.7% | 27.8% | 93.4% [91.5%, 95.1%] | 8 |
| gpt-oss-safeguard 20B | system-role policy | 113 | 43.5% | 12.0% | 100% [100%, 100%] | 16 |

structure that rebinds depends on the guard. On Llama Guard 3 every syntax works (41 of 52 attack families at $\geq 70\%$), while on Qwen3Guard only the full-task forgery does and generic labels stay at or below 18.3% (see "What counts as a task" below). The decoy's harm matters on four guards: with a harmful decoy the strongest forgeries pass $\leq 13.6\%$ on Shieldstral, $\leq 11.1\%$ on Llama Guard 3, 0% on Qwen3Guard and $\leq 9.4\%$ on gpt-oss-safeguard.

Two guards depart from the Shieldstral pattern. Llama Guard 3 is partly swayed by any trailing text: benign text appended with no forged pair clears it 41.2% [36.1%, 46.0%] (Shieldstral 4.3%). Part of its bypass is recency, although the forged pair still adds up to 59 points. On ShieldGemma bypass barely depends on the decoy (the forged triple of instruction, query and document clears 84.3% with a harmful decoy and 83.3% with a benign one), which is closer to generic evasion than to rebinding. ShieldGemma's best family, like Qwen3Guard's, places the forgery in the middle position, inside benign filler; appended alone, the guard-native forgery reaches 73.6% and 81.7% on them.

**gpt-oss-safeguard.** The base guard is a weak detector on this data: it flags 56.5% [49.6%, 63.2%] of harmful core items and 12.0% [8.2%, 17.2%] of benign ones, and bypass is measured on the 113 it flags. The forged triple clears 100%, but its instruction field is the directive $S$, and it therefore does not separate rebinding from persuasion. The guard-native forgery, with no instruction sentence, clears 71.7% [65.2%, 78.5%] with one pair (multi-verdict replies are counted as flagged, Section C).

**What counts as a task.** Our hypothesis, consistent with the results but not established causally, is that each guard grades the most recent complete *grading task* in its input (the developer-written fields it answers, plus a document or conversation), and that its format sets what counts as complete. On Shieldstral a query–document pair is a complete task (Table 1). On gpt-oss-safeguard a forged pair in a generic syntax gets through only 2–13% of the time, far less often than the guard-native forgery, which opens new developer and user turns (above). In an exploratory probe on 40 flagged train-split items of Qwen3Guard, forging only the conversation clears 5/40 (13% [5%, 26%]), while forging the whole task (the template's closing instruction tail, then a new task line, policy, category list and conversation) clears 40/40 (100% [91%, 100%]). With a harmful decoy the same block stays flagged (median 0.997), and on the test core set the length-matched filler clears Qwen3Guard only 4.7% [3.0%, 6.5%]. Length alone does not explain the effect. A defense must therefore protect the whole task (instruction, policy and content), not only the content delimiters or the conversation markers.

## 5 Defense: Boundary Tokens and Segment-Max Training

Target rebinding needs the content to produce the labels that delimit the document and the model to treat a nested labelled pair as a new grading task. Our defense removes both preconditions: Part B (boundary) marks the document with tokens that content cannot produce, and Part T (training) teaches the guard to grade everything inside that region as one document and to score it at least as high as its most harmful part. We compare {none, B, T, B+T} with training-free baselines (Section 6).

**Part B: a boundary the content cannot produce.** We build the *bounded format* by rendering the guard's *stock format*, its unmodified chat template, with an empty document slot, which leaves the trusted prefix and suffix as they are. We tokenize the content on its own with special-token parsing disabled, and any control-token string inside it becomes ordinary text tokens. Disabling special-token parsing affects only tokens marked special, and an added token that is not marked special is still emitted as one id [44, 47]. On Shieldstral, the guard we train the defense on, no added or special id is reachable from content under this encoder (Section D). We then wrap the content's token ids in two reserved *boundary tokens* that open and close the document. Both ids are unused in pretraining, and no input text tokenizes to them. We remove nothing from the content, so a literal `<Query>:` or `[/INST]` reaches the model as plain text inside the boundary. This is the reserved-delimiter idea of StruQ and SecAlign [4, 5], applied to a classifier's content slot. B alone, with untrained boundary embeddings, tests whether the format change helps without training.

**Part T: segment-max self-distillation.** T trains the *student*, the guard being trained, to score the whole content at least as high as its most harmful part. The *teacher* $f_0(\cdot)$ is the frozen base guard scoring a text on its own as $P(\text{yes})$, in the stock format with no attack around it. For content $c$ with segments $S(c)$,

$$\text{target}(c) \;=\; \max\Big(f_0(c),\; \max_{s \in S(c)} f_0(s)\Big). \tag{1}$$

In an attacked training example the segments are the original request and the decoy. Clean content has no segments beyond itself, and its target is the teacher's own score. The whole-content term keeps the target at least as high as the base guard's score on the whole content, which covers harm that appears only in combination when the base guard sees it [50]. The max is defined without knowing which segment is the injection. Existing guidance already scans long inputs in segments at inference [32], and we run that as a baseline (Section 6). Here the max is a training target: the defended guard reads the content in one pass and needs no segmenter. When the decoy is benign, the target reduces to clean-context distillation [12, 38]. A *clean-copy target*, in which the teacher reads only the original item, is an ablation that tests whether the max matters. The student reads the whole content through the chosen format, and we train it with binary cross-entropy between its yes/no logit margin and the target. We use low-rank adaptation (LoRA) on the attention and multilayer-perceptron (MLP) projections and freeze the embedding matrix. In B+T we also train the two boundary-token embedding rows jointly, and T alone uses the stock format (Section D).

**Training data.** We generate the training attacks and therefore know their segments. Each of 5,000 train-split harmful or benign items appears once clean and twice with attacks sampled from the training attack families, and 1,250 *structured-benign* items, benign text with label-like structure, bring the total to 16,250 examples. Each attack draws its decoy and forged query, per example, from the training half of the matching pool (Section D). The structured-benign items are *hard negatives*, generated from nine templates (FAQs, forms, chat logs, markdown, JSON and others) filled with benign train-split text. Their targets come from the teacher, and the model learns from them that structure alone is not a reason to flag, in the spirit of NotInject/InjecGuard [20]. The structured-benign test set uses the same templates with test-split fill text. Over-refusal on it measures new

content in familiar templates. Training sees no test item, no held-out attack family (Section B) and no decoy or forged query from the held-out halves of the pools.

# 6 Evaluating the Defense

## 6.1 Setup

The main model is Shieldstral-1.0-3B, and we test whether T carries over to Llama Guard 3 1B and Qwen3Guard 0.6B [14, 48]. All numbers are on the test split, and no test item is seen in training. We measure bypass at $\tau = 0.5$ on the harmful test items that each condition flags with no attack. The attack sets are the training families with each family's fixed decoy and query, the held-out families (whole grammar branches never seen in training, Section B) with the same fixed decoys and queries, and the held-out families with decoys and forged queries drawn from the held-out halves of the pools (three instantiations per item and family, seeded by the item). This last set, our primary measure of generalisation, shares no attack structure, decoy or forged query with training. Position is not held out: every sampled form in both family splits appends the forged structure after the request, and only the structural and score-guided searches vary position (Section 6.3). Bypass pools the sampled attacks whose decoy is benign or borderline, since an attack whose decoy is harmful is correctly flagged.

*Detection* is the area under the ROC curve (AUROC), recall and false-positive rate (FPR) at $\tau$ on one fixed random sample of 1,500 clean test items (801 harmful, 699 benign), shared by all conditions. *Over-refusal* is the flag rate on XSTest-safe [39] ($n = 75$), OR-Bench-hard [10] ($n = 400$) and a structured-benign set ($n = 500$). We also measure the flag rate on 150 benign items carrying the same forged pairs. T has two seeds (we report the range), and B+T, the clean-copy target and the generality runs have one. This seed plan (cut from the pre-registered three seeds for B+T to save compute), the held-out-decoy design, the score-guided search and the two training-free baselines come from a protocol deviation recorded after a first round of test-split runs, with those results in view (deviation D4, Section A). Bypass differences use a paired bootstrap over items. Detection and over-refusal differences use unpaired 95% intervals, which are wider than paired ones, and a condition counts as non-inferior only if that interval lies inside the pre-registered margin.

**Baselines.** Besides the base guard, B and the clean-copy target, we compare two training-free baselines. The *fail-closed structure detector* flags any content that contains a chat control-token string or a label-like string from a fixed English list of syntaxes, and passes everything else to the base guard. *Inference-time segment-max* splits the content at the same strings, scores every segment and the whole content with the base guard, and returns the maximum, which is the target of Equation (1) computed at run time, at the cost of one forward pass per segment. In the confirmatory study, an anti-injection directive (a sentence in the instruction telling the guard to disregard instructions inside the document) left bypass unchanged, and escaping and fencing left it at 88–93% (Section A).

## 6.2 Main results

Training cuts bypass on held-out families with held-out decoys and queries from 75.0% [74.0%, 76.0%] of attacked inputs on the base guard to 0.8% and 1.6% on T (seeds 0 and 1) and 0.9% [0.7%, 1.1%] on B+T, meeting the pre-registered target of $\leq 10\%$ (Table 3). Held-out decoys clear T slightly less often than the fixed ones (by 0.4 points [+0.1, +0.8], paired, $n = 290$ items), and on the training families with held-out decoys T is cleared on 0.7–1.4% (base 69.4%). B alone leaves held-out bypass at 76.8%, as pre-registered: untrained boundary embeddings mean nothing to the model. T also generalises worse than B+T to held-out families, as predicted (paired-bootstrap T − B+T = +0.7 points [−0.1, +1.5], one-sided $p = 0.04$, $n = 290$ items), but the effect is under one point and both conditions are near 1%.

Table 3: Defense on the test split, Shieldstral, $\tau = 0.5$, in %. Bypass is over benign or borderline decoys (lower is better): training and held-out families with the fixed decoy and query, and held-out families with held-out decoys and queries, the primary measure. Detection is on the clean test sample. T has two seeds (range); the other trained rows have one. Intervals for the held-out-decoy column are in the text.

| | bypass ↓ | | | detection | | |
|---|---|---|---|---|---|---|
| | fixed decoy | | held-out | | | |
| condition | train | held-out | decoy | AUROC | recall | FPR |
| none (base) | 87.9 | 85.1 | 75.0 | 0.810 | 47.1 | 5.9 |
| B (untrained boundary) | 90.7 | 85.4 | 76.8 | 0.827 | 47.8 | 5.4 |
| structure detector | 0.0 | 24.5 | 23.2 | 0.814 | 47.9 | 5.9 |
| inference-time segment-max | 0.0 | 24.6 | 23.4 | 0.810 | 47.1 | 5.9 |
| T (2 seeds) | 1.0–1.8 | 1.2–2.0 | 0.8–1.6 | 0.802–0.803 | 46.4–46.7 | 5.2–5.3 |
| B+T | 1.2 | 1.0 | 0.9 | 0.800 | 46.9 | 5.3 |
| T, clean-copy target | 2.0 | 2.2 | 2.1 | 0.817 | 47.4 | 5.6 |

Table 4: Utility and trailing-content checks for the conditions of Table 3, in %. Over-refusal is the flag rate on benign sets (lower is better). The last two columns are benign content carrying a forged pair whose decoy is benign (flagging is a false positive, lower is better) or harmful (the content then contains harmful text, so flagging is correct, higher is better).

| | over-refusal ↓ | | | benign + forged pair | |
|---|---|---|---|---|---|
| condition | OR-Bench-hard | XSTest-safe | structured benign | benign decoy ↓ | harmful decoy ↑ |
| none (base) | 51.5 | 6.7 | 5.8 | 0.2 | 100 |
| B (untrained boundary) | 52.8 | 8.0 | 6.8 | 0.1 | 100 |
| structure detector | 51.5 | 6.7 | 58.6 | 87.5 | 100 |
| inference-time segment-max | 51.5 | 6.7 | 9.8 | 13.0 | 100 |
| T (2 seeds) | 52.0–54.3 | 2.7–5.3 | 5.0 | 0.7 | 100 |
| B+T | 51.8 | 4.0 | 5.8 | 0.8 | 100 |
| T, clean-copy target | 50.8 | 5.3 | 5.2 | 0.7 | 0.8 |

Measured costs in detection and over-refusal are small, but non-inferiority is not established (Table 4). Relative to the base guard, B+T changes detection AUROC by $-0.010$ and recall by $-0.1$ points, and no over-refusal rate rises by more than 0.25 points (one OR-Bench-hard item). These point estimates lie inside the pre-registered non-inferiority margins, but the 95% intervals of the differences are wider, and non-inferiority is confirmed for neither detection nor over-refusal (intervals in Section A). T's changes are of the same size, with one exception: T seed 1 exceeds the over-refusal margin on OR-Bench-hard (+2.75 points against ±2). Benign content carrying a forged pair with a benign decoy shows a small but statistically detectable cost: it is flagged 0.7% of the time under T (both seeds) and 0.8% under B+T, against 0.2% for the base guard (Fisher exact $p = 0.012$ for each T seed, $p = 0.003$ for B+T).

**Baseline results.** The training-free baselines stop every training-family attack but let through 23.2% [22.2%, 24.2%] (detector) and 23.4% [22.4%, 24.4%] (inference-time segment-max) of held-out families with held-out decoys. Both split or match content with a fixed English pattern list, and all of their remaining bypass comes from the two held-out families with non-English labels: one with Spanish labels in markdown-bold syntax and one with Chinese labels, which clear the detector 97.2% and 88.4% of the time. Every other held-out family clears it 0%. A pattern list that included those labels would flag both families, so the detector's held-out number reflects how well its list covers the attacker's labels. The detector's cost is false positives: it alone flags 54.8% [50.4%, 59.1%] of the structured-benign test set (58.6% with the guard's

Table 5: Adaptive attacks on Shieldstral, 50 flagged test items, seed 0. Each cell is the number of items cleared, with Wilson 95% interval. "All attacks" is the per-item worst case over the score-guided search, the structural search and GCG (run on 20 of the 50 items, and not on B+T, so the B+T cell is a lower bound). The plain-text control adds benign user prompts with no forged structure and is not counted in "all attacks".

| guard | score-guided search | structural search | all attacks | plain-text control |
|---|---|---|---|---|
| none (base) | 50/50 [93%, 100%] | 50/50 [93%, 100%] | 50/50 [93%, 100%] | 35/50 [56%, 81%] |
| T | 47/50 [84%, 98%] | 4/50 [3%, 19%] | 49/50 [89%, 100%] | 26/50 [39%, 65%] |
| B+T | 34/50 [54%, 79%] | 4/50 [3%, 19%] | ≥34/50 (no GCG) | not run |

own flags) and 87.5% of benign content carrying a forged pair. Inference-time segment-max over-flags less than the detector but more than T: it flags 9.8% of structured-benign content (T: 5.0%) and 13.0% of benign content carrying a forged pair with a benign decoy (T: 0.7%).

**Naive adversarial training.** Naive adversarial training, a LoRA on the same examples with hard 0/1 ground-truth labels, matches T on bypass but costs clean FPR and misses trailing harmful content. It was trained and tested only under the fixed-decoy protocol (Table 7, two seeds), not with held-out decoys, and it reaches 0.9–1.1% bypass on held-out families (T: 1.2–2.0%) and 0.0% structured-benign over-refusal, on a set that overlaps its hard negatives, which contradicts our pre-registered prediction that it would cost at least 3 points there or generalise worse. Its clean FPR is 12.6–14.9% (T: 5.2–5.3%), and its higher AUROC (0.921–0.925) comes from being the only condition trained on ground-truth labels, not the base guard's scores.

**Trailing harmful content.** The max in the target keeps the trained guard reading past a forged pair. A guard trained against rebinding could stop reading there [43] and so open a smuggling channel, which the harmful-decoy column of Table 4 tests. T and B+T flag 100% of these items with the fixed decoys and 95.5–98.7% with held-out ones, against 0.8% (clean-copy target) and 8.0–8.6% (naive adversarial training).

**Other guards.** T, trained and tested with the fixed decoy and query (Table 7), transfers to the two smaller guards without changing the recipe. On Llama Guard 3 1B it cuts held-out-family bypass from 82.0% [80.4%, 83.5%] to 2.5% [1.9%, 3.2%]; on Qwen3Guard 0.6B the sampled families are mostly generic syntaxes it does not rebind to (Section 4.2), and the drop from 9.5% to 0.0% says little. On both, the flag rate on benign content carrying a benign forged pair rises (from 2.4% to 22.5% on Llama Guard 3 and from 10.5% to 25.9% on Qwen3Guard), for reasons we do not know.

### 6.3 Adaptive evaluation

The defense resists enumeration of the attack grammar, but an attacker with score access or gradients still breaks it. The attacker knows the defense and sees its score (Section 2), and we follow the evaluation norms of Carlini et al. [2]. The content cannot produce the boundary tokens (Section 5). The attacker therefore optimises the content text only, and every attack runs end to end on the defended model. We run three attacks on 50 flagged test items and report the per-item worst case (Table 5). *Score-guided search* hill-climbs on the defended guard's score within 300 guard calls per item, mutating one or two fields of the best candidate (syntax, structure, position, number of pairs, label strings, decoy text and forged query), with random restarts. Its decoys are the held-out halves of the decoy pools plus 500 benign user prompts from the development split, and its queries are every forged-query pool plus the deployment's own query; most of this material is unseen in training. *Structural search* tries all 654 forms of the full attack grammar and 30 random label mutations,

with decoys and queries drawn from the held-out pools. *GCG* (Greedy Coordinate Gradient [51]) optimises a 20-token suffix inside the forged document for 250 steps, on 20 of the 50 items. A *plain-text control* runs the score-guided search with the same budget and decoy texts, but each candidate adds only the decoy text (one to five copies, before or after the request), without labels or forged query.

Both searches clear the undefended guard on every item, passing the sanity check of at least 90%. Enumerating the grammar with held-out decoys and queries rarely beats the trained guards: it clears T on 8% [3%, 19%] of items and B+T on 8%, within our pre-registered target of at most 20% for B+T.

The score-guided search breaks both trained guards, clearing T on 94% [84%, 98%] of items and B+T on 68% [54%, 79%]. Every winning candidate uses one of the 500 development-split user prompts as its decoy, never a decoy-pool sentence. Alone, the 35 distinct winning prompts are benign to the base guard (median score 0.0003; one, in one B+T win, scores 0.53). In 34 of T's 47 wins the best candidate places the forged structure before the request, often with an off-topic forged query (21 of the 34), and on B+T 29 of 34 wins prepend. One interpretation is that prepended structure attacks the grading question and not the document, since almost all training attacks append the pair after the request (244 of the 252 training forms). The search does not separate this from the shift to longer, prompt-like decoys. With one seed per guard, B+T is cleared less often than T.

The plain-text control measures how much of this needs forged structure. On 26 of the 47 items the search wins against T, benign user prompts with no labels and no forged query, chosen by the same score-guided search, also clear T, and forged structure adds the other 21. The control clears the base guard on 70% [56%, 81%] of items and T on 52% [39%, 65%], always with the benign text before the request. The inert filler of Table 1 (0%) is one fixed text, whereas the control picks, per item, the best of up to 300 benign candidates using the guard's score. Each item's first candidate is drawn before any score is seen, and it alone clears the base guard on 14 of 50 items (28% [17%, 42%]) and T on 3 (6% [2%, 16%]): an unsearched benign candidate does dilute the request, but selection under score access accounts for most of the control's success.

GCG clears T on all 20 items it was run on, which raises the per-item worst case over all attacks against T to 49/50 (98% [89%, 100%]). We did not run GCG on B+T, so its worst case, 34/50, is a lower bound.

## 7 Related Work

**Structure forgery against generators and judges.** Liu et al. [25] formalise prompt injection against LLM-integrated applications. Their "fake completion" and "combined" attacks forge the end of one task and the start of another, and we apply the same construction to guards. In attacks descended from indirect prompt injection [15], forged template or conversation structure redirects agents and chat models [3, 8, 11, 23, 45], and injected content can manipulate the scores of LLM judges [21, 40, 41]. Those targets are generators that carry out the injected task, or judges whose answer shifts. Ours are safety classifiers that grade the forged task, so a successful attack is a false "safe" verdict on content that still contains the harmful request.

**Attacks on guards.** Guards have been bypassed with universal perturbations [26], character-injection evasion [16] and pipeline attacks [30], and a forged judge-output block bypassed OpenAI's LLM-based detectors [29]. The closest work is concurrent: Li et al. [22] forge verdict and authority fields in a typed judge's structured input, and none of their forgeries introduces a new task to grade. Our content forges a task instead of a verdict, and we also train a defense and test it against adaptive attackers.

**Structured-query and distillation defenses.** StruQ and SecAlign [4, 5] combine reserved delimiters with training to ignore injected instructions. Meta-SecAlign [7], DefensiveTokens [6] and learnable trust-boundary delimiters [49] extend this line, and SecOPD [38] and the concurrent RAISED [12] distill the model toward its behaviour on clean context. All of them defend generative instruction-followers. We apply the idea to a

classifier's content slot, where the output is one score, and our segment-max target does not need to know which segment is the injection. Fine-tuning defenses that report near-zero attack success have been broken by adaptive attacks [34, 36], and ours is broken the same way (Section 6.3).

# 8 Limitations and Conclusion

**External validity.** The confirmatory results (Section 3) cover one guard and one corpus (AdvBench rows 200–299), in English, on short text-only documents, and on the broader benchmark corpus the same pair clears Shieldstral less often (Section 3.1). The cross-model results are label-only and use 30 AdvBench requests, and for Llama Guard 4 we did not separate rebinding from a malformed chat template (Section 4). gpt-oss-safeguard's bypass rates cover the 113 items it catches, one unpinned API route (OpenRouter, served by Groq) and the user-message channel, and we cannot observe how the provider tokenizes turn markers sent as text. Every bypass rate is conditional on the items the guard flags with no attack, and Shieldstral's clean recall is 47% on the test split, with a miss rate of 50% on the benchmark core set. The templates of Llama Guard 3 and Qwen3Guard ask for an assessment of the last user turn only, and on these guards rebinding is as much a gap in the task specification as a parsing failure. We did not measure whether a downstream model acts on the forwarded request, so bypass is an upper bound on end-to-end harm. Retrieval, tool-result and multi-turn channels were not measured, and model size was not varied within a series.

**Evidence strength.** The Qwen3Guard probe of what counts as a task is exploratory (40 train-split items, one decoy), and the activation patching (Section C) shows where the verdict is not carried but not which components choose the graded document. T has two seeds on Shieldstral, while B+T, the clean-copy target, the adaptive attacks and the smaller guards have one. The adaptive attacks use 50 items (20 for GCG), GCG was not run on B+T, and we did not run the pre-registered LLM-based attacker. The smaller guards and naive adversarial training were evaluated only with one fixed decoy and query per family, so the comparison of naive adversarial training with T does not use the held-out-decoy measure.

**What the defense does not cover.** Our held-out results cover appended forms only. Training closed the appended channel, and the score-guided search found a prepended one, although the search does not separate prepended structure from its decoy distribution (Section 6.3). The defense does nothing against the benign prompts of the plain-text control. On the smaller guards it raises over-refusal on benign content carrying forged structure to 22–26%. Segment-max can miss harm that exists only when benign segments are combined. Scoring the whole content as one more segment reduces this risk without removing it; we ran no composition attack, and the harmful-decoy test covers only the orderings our grammar and search generate. We do not claim general prompt-injection robustness, which Pant et al. [37] argue is unattainable in shared-embedding models without enforced separation.

**Conclusion.** Guards that read content through the plain-text channel that states their task grade the last complete task they see, so content that forges one gets a harmful request approved. Training the guard against the maximum over the content's parts closes this channel for held-out appended attacks but not against an adaptive attacker, whose score-guided search still gets through. Deployers should give the content region a boundary the content cannot reproduce, tokenized with special-token parsing disabled and no added tokens outside the special set [44, 47], and should not let an agent act on a guard verdict alone. Researchers who build defenses against rebinding should report the plain-text control alongside them.

## 9 Ethics and Disclosure

**Disclosure.** We reported the attack to the vendors of the affected guards before writing this paper, and none had responded at the time of writing. An informal version of the attack has been public since an earlier blog post [19], so we release the paper, the benchmark and the defense without an embargo. Vendors can adopt the defense as a mitigation.

**Harms and benefits.** The attack is simple and transfers across guards, so we document it and offer a fix instead of leaving it latent. We author no harmful text. Every harmful request, and every short example we reproduce, comes from published benchmarks (Section 4).

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# A Protocol and Pre-registered Outcomes

We froze the first protocol before the confirmatory run and did not edit it after seeing results, and we record every deviation from either protocol. These runs are exploratory and fall outside both frozen protocols: the wider set of 100 requests, the three-decoy comparison, the softened-query control, the forged-query (attacker knowledge) variants, the forged-verdict and split-payload controls, the real-control-token probe, the colon-label detector and label-normalisation tests, the clean-copy ablation and the GCG control without a forged pair. The second protocol was frozen before any test-split run and has four recorded deviations.

- D1 adds the variability design: three instantiations per item and family, two paraphrased deployment policies on six families, and three training seeds per defense condition.
- D2 replaces Qwen3Guard's conversation-only native forgery with the full-task forgery found on the train split, and adds the length control.
- D3 trims compute: 5,000 training items with two attacks each, three seeds for B+T and two for T and naive adversarial training, GCG on 20 items at 250 steps, and structural search on 50 items.
- D4 was recorded after a first round of test-split defense runs and before the reported ones, so it is a deviation made with test results in view. The fixed-decoy conditions of Table 7 come from that first round. D4 draws training decoys and forged queries from one half of each pool and evaluates on the other half, adds the score-guided search and the two training-free baselines, and replaces the D3 seed plan with a smaller set to save compute: two seeds for T and one for B+T and the clean-copy target.

Table 6 lists every prediction with its outcome.

Table 6: Pre-registered predictions and outcomes. P: confirmatory study; M: benchmark; F, A: defense. Intervals are 95%.

| | prediction | outcome |
|---|---|---|
| P1 | instruction-only $\leq 10\%$ | confirmed (0%) |
| P2 | forged pair $\geq 70\%$ | confirmed (100%) |
| P3 | harmful decoy lowers bypass | not a valid test as run: the panel's "harmful decoy" scores 0.03 alone, so it is the borderline decoy; bypass fell to 29–35%; the exploratory three-decoy comparison and M3 address the prediction |
| P4 | not recency | confirmed (append 100%, prepend 15%) |
| P5 | syntax not load-bearing | confirmed (all variants high, no labels 0%) |
| P6 | wide policy-dependence | refuted: median 99%, range 87–100% |
| P7 | directive does not fix | confirmed (unchanged) |
| P8 | cross-model transfer | confirmed (30/30, 29/29) |
| P9 | escaping fixes it | refuted: bracket-escape 88%, label-strip 91%, nonce-fence 93% |
| M1 | a core family $\geq 70\%$ on every guard | confirmed; best family 93.4–100% |
| M2 | instruction-only $\leq 10\%$ on every guard | not confirmed; $\leq 7.0\%$ on four guards, 20.2% on ShieldGemma |
| M3 | dose–response, $\rho \leq -0.8$ and non-increasing bins (Shieldstral, Qwen3Guard) | partly supported on Shieldstral: $\rho = -0.84$, interval $[-0.93, -0.67]$ over decoys ($[-0.85, -0.82]$ over requests), $-0.78$ on the dev split, bins not monotone; not run on Qwen3Guard |
| M4 | control-token syntax $\geq$ angle; gap closes with special parsing off | partly supported; higher in 7 of 8 comparisons but clearly so only on Shieldstral; the parsing-off arm was not part of the benchmark run |
| M5 | exploratory: category, source, size | released with the artifacts; size not varied within a series |
| F1 | B alone leaves held-out bypass $\geq 50\%$ | confirmed; 76.8% [75.8%, 77.8%] |
| F2 | T generalises worse than B+T on held-out families | confirmed by the pre-registered criterion, with an effect under one point; T − B+T = +0.7 points $[-0.1, +1.5]$, one-sided $p = 0.04$ ($n = 290$) |
| F3 | B+T held-out bypass $\leq 10\%$ | confirmed; 0.9% [0.7%, 1.1%] |
| F4 | B+T AUROC within 0.01, recall within 2 points (on the interval of the difference) | not confirmed; AUROC $-0.010\,[-0.041, +0.021]$, recall $-0.1\,[-5.0, +4.8]$: point estimates inside the margins, intervals wider |
| F5 | B+T over-refusal within 2 points (on the interval of the difference) | not confirmed; XSTest-safe $-2.7\,[-11.1, +5.4]$, OR-Bench-hard $+0.25\,[-6.6, +7.1]$, structured benign $0.0\,[-3.0, +3.0]$: no point estimate rises by more than 0.25, intervals wider than the margin |
| F6 | naive adversarial training costs $\geq$ 3 points structured-benign over-refusal or generalises worse | not confirmed with the fixed decoy (0.0% on a set overlapping its hard negatives, 0.9–1.1% held-out); its costs are clean FPR and the trailing-pair blind spot |
| A1 | GCG against B+T $\leq 30\%$ | not tested on B+T; on T GCG clears 20/20 |
| A2 | structural search against B+T $\leq 20\%$ | confirmed; 4/50 (8% [3%, 19%]); the score-guided search, added in D4 without a prediction, clears 34/50 |
| A3 | LLM attacker | not run |

# B Benchmark Grammar

The attack grammar has the axes below. The full grammar yields 659 forms (654 attacks, 5 controls), and the core panel has 63. A family is held out if its syntax is JSON, YAML, markdown bold or control token, if its language is neither English nor Spanish, or if it has at least three forged pairs. Spanish-label families are therefore held out only through the syntax or pair-count rule. The defense is trained on the remaining families.

- label syntax: guard-native tags, angle brackets, square brackets, markdown, plain colon, abbreviations, JSON, XML, YAML, control tokens;
- number of forged pairs: 1–5; position: append, prepend, middle;
- decoy harm: benign, borderline, harmful; forged query: exact, paraphrase, generic, out-of-domain; language; encoding: plain text or real control tokens.

Per-family, per-category and per-source bypass and threshold sensitivity are released with the artifacts.

# C Additional Attack Analyses

**Activation patching.** Activation patching on Shieldstral is consistent with rebinding but does not identify the circuit that implements it. We run the attacked input and the same input with request and decoy swapped ($n = 100$), using for each request a decoy of the closest token length and matching positions by role, not by index, which keeps any remaining length mismatch from misaligning them. Patching the residual stream at the final position flips the verdict on at least 99% of inputs at every layer from 20 onward (of 26, numbered from 0). Patching the forged labels, the forged query or the decoy's positions at any layer moves the yes/no logit margin by at most 0.05 of the gap between the two inputs and flips at most 3% of verdicts.

**Benchmark variations.** Real control-token syntax changes bypass relative to angle-bracket labels by $-1.5$ to $+10.4$ points, which only partly supports our prediction that it would raise bypass. Across the three deployment policies, bypass moves by at most 6.8 points on Shieldstral, Llama Guard 3 and Qwen3Guard, and by up to 13.7 points on ShieldGemma.

**Multiple verdicts from gpt-oss-safeguard.** On stacked forged pairs gpt-oss-safeguard sometimes returns one verdict per content block. We count an item as flagged if any verdict is 1, the reading least favourable to the attack, chosen after the multi-verdict replies were seen. Reading only the last verdict, as a pipeline acting on the trailing label would, raises two stacked-pair families from 2.9–3.5% to 87.3–90.6% and moves every other family by under one point.

# D Defense Implementation and Training

**Bounded format.** The bounded format renders the stock chat template with a placeholder document and splits it into a trusted prefix and suffix. The content is encoded on its own with special-token parsing disabled, and the two reserved boundary tokens (unused in pretraining) are inserted around it. The template puts a space before the document, which the tokenizer would otherwise merge into the content's first token. We move that space into the trusted prefix, so plain documents tokenize exactly as they do in the stock format (verified on 1,600 documents: AdvBench rows 200–299 and the clean test sample).

**Training.** We train a LoRA (rank 16, $\alpha = 32$) on the language model's attention and MLP projections with AdamW at learning rate $10^{-4}$, batch 8, one epoch and maximum length 384. The loss is binary cross-entropy between the yes/no logit margin and the segment-max teacher target. LoRA does not touch the embeddings. In B+T a trainable two-row offset, initialised at zero, is added to the embeddings of the two boundary tokens (ids 36 and 37), and every other embedding stays frozen. The data are 5,000 train-split prompt items, each clean and with two training-family attacks, plus 1,250 structured-benign items, for 16,250 examples. One epoch is 2,032 steps, the last batch partial. Each attacked example's decoy and forged query are a seeded draw from the training halves of the pools: 20 of the 40 benign decoys, 3 of 6 borderline and 4 of 8 harmful decoys, and 4 of the 8 forged queries at each relatedness level. The hard negatives are train-split structured-benign items. Naive adversarial training and the clean-copy ablation use the same examples with different targets.

**Compute and adaptive budgets.** Training and the test-split defense, adaptive and mechanism runs used one NVIDIA RTX 4090. A training run takes about 17 minutes (1,028 s for T and 1,037 s for B+T at the last logged step, 2,030 of 2,032). The benchmark ran on an Apple-silicon laptop GPU. The score-guided search has 300 guard calls per item on 50 items, starting from 20 random candidates and then mutating batches of 10 children with restart probability 0.2. The structural search scores all 684 attacks on each of 50 items (batch 128). GCG optimises a 20-token suffix inside the forged document for 250 steps (batch 32 candidates per step) on 20 items, and stops on an item at its first success.

**Fixed-decoy conditions.** Table 7 gives the conditions trained and tested with one fixed decoy and query per family: naive adversarial training and the generality runs of T on Llama Guard 3 1B and Qwen3Guard 0.6B (Section 6). They have no held-out-decoy column. The hard negatives of the trained rows include 380 of the 500 structured-benign test items, so their structured-benign column is not a held-out measure.

Table 7: Fixed-decoy conditions on the test split, $\tau = 0.5$, in %. Columns as in Tables 3 and 4; naive adversarial training has two seeds (range), the other rows seed 0.

| | bypass ↓ | | detection | | | over-refusal ↓ | | | benign + forged pair | |
|---|---|---|---|---|---|---|---|---|---|---|
| condition | train | held-out | AUROC | recall | FPR | OR-B. | XST. | struct. | benign | harmful |
| naive adv. training (2 seeds) | 0.9–1.0 | 0.9–1.1 | 0.921–0.925 | 83.3–84.6 | 12.6–14.9 | 0.3–0.8 | 8.0–10.7 | 0.0 | 7.8–8.2 | 8.0–8.6 |
| Llama Guard 3 1B, base | 84.0 | 82.0 | 0.758 | 58.7 | 16.9 | 40.5 | 1.3 | 23.4 | 2.4 | 97.3 |
| Llama Guard 3 1B, T | 2.3 | 2.5 | 0.771 | 58.4 | 15.0 | 38.3 | 1.3 | 23.0 | 22.5 | 100 |
| Qwen3Guard 0.6B, base | 28.3 | 9.5 | 0.917 | 88.5 | 23.6 | 76.8 | 21.3 | 30.4 | 10.5 | 100 |
| Qwen3Guard 0.6B, T | 0.0 | 0.0 | 0.924 | 89.0 | 23.3 | 77.3 | 20.0 | 29.4 | 25.9 | 100 |

OR-B.: OR-Bench-hard; XST.: XSTest-safe; struct.: structured benign; benign / harmful: decoy of the forged pair on benign content.

**Content tokens.** Disabling special-token parsing leaves atomic any added token that the tokenizer does not mark special [44, 47], so we checked which added-vocabulary ids the content encoder can still emit. For each guard tokenizer we encoded every added token's string on its own, embedded in plain text and in 500 random concatenations, with special-token parsing disabled, and recorded any added or special id in the output. No added or special id is reachable from content on Shieldstral, the guard the defense is trained on (1000 added tokens, all special, with the boundary tokens at ids 36 and 37), or on Llama Guard 3 1B (256, all special). On Qwen3Guard-0.6B, 12 of 26 added tokens are not marked special and stay atomic, among them the tool-call tags <tool_call> and </tool_call> and the fill-in-the-middle markers. On ShieldGemma 2B, 242 of 249 are not special (the <unused$k$> and other reserved markers) and also stay atomic. Content can produce these ids, so a boundary-token design on these tokenizers must not use them as boundary tokens.

# E Reproducibility

The runs used these Hugging Face revisions: Shieldstral-1.0-3B `003ec7e`, Llama Guard 3 1B `acf7aaf`, Qwen3Guard-Gen-0.6B `fada3b2` and ShieldGemma 2B `d1dffc9`. Local guards are scored from a single forward pass with no sampling. The score is the probability of the unsafe label renormalised over the guard's label tokens, computed in fp32 from bf16 hidden states. We reached the API-served guards, gpt-oss-safeguard-20B and Llama Guard 4 12B, through OpenRouter at temperature 0 without pinning a provider. Every benchmark call to gpt-oss-safeguard was served by Groq, as recorded in the response metadata; the cross-model runs did not record the provider. Dataset sources and licenses: AdvBench, HarmBench, Aegis 2.0, BeaverTails, the OpenAI moderation evaluation and ToxicChat, with XSTest and OR-Bench as over-refusal sources. We use the non-commercial sets (ToxicChat, BeaverTails) for research only. All hyperparameters, splits and the full protocol are in Sections A and D.

# F Open Science

We release the GuardRebind benchmark and harness, the attack grammar, the defense code, the training commands and data recipe, and the result files behind every number in this paper, at `https://github.com/ukanwat/guard-policy-injection`. The released results contain item ids and scores, plus text only from the public benchmarks we used (such as the AdvBench requests of the confirmatory run and the benign development-split prompts that won as search decoys) and model outputs that quote it. We release no new harmful content. The pre-registered protocol and its deviations are included so that predictions can be checked against outcomes.